

TABLE 3.3 One-dimensional, steady-state solutions to the heat equation with no generation

	Plane Wall	Cylindrical Wall ^a	Spherical Wall ^a
Heat equation	$\frac{d^2T}{dx^2} = 0$	$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$	$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$
Temperature distribution	$T_{s,1} - \Delta T \frac{x}{L}$	$T_{s,2} + \Delta T \frac{\ln(r/r_2)}{\ln(r_1/r_2)}$	$T_{s,1} - \Delta T \left[\frac{1 - (r_1/r)}{1 - (r_1/r_2)} \right]$
Heat flux (q'')	$k \frac{\Delta T}{L}$	$\frac{k \Delta T}{r \ln(r_2/r_1)}$	$\frac{k \Delta T}{r^2 [(1/r_1) - (1/r_2)]}$
Heat rate (q)	$kA \frac{\Delta T}{L}$	$\frac{2\pi Lk \Delta T}{\ln(r_2/r_1)}$	$\frac{4\pi k \Delta T}{(1/r_1) - (1/r_2)}$
Thermal resistance ($R_{t,\text{cond}}$)	$\frac{L}{kA}$	$\frac{\ln(r_2/r_1)}{2\pi Lk}$	$\frac{(1/r_1) - (1/r_2)}{4\pi k}$

^aThe critical radius of insulation is $r_{cr} = k/h$ for the cylinder and $r_{cr} = 2k/h$ for the sphere.

3.5

Conduction with Thermal Energy Generation

In the preceding section we considered conduction problems for which the temperature distribution in a medium was determined solely by conditions at the boundaries of the medium. We now want to consider the additional effect on the temperature distribution of processes that may be occurring *within* the medium. In particular, we wish to consider situations for which thermal energy is being *generated* due to *conversion* from some other energy form.

A common thermal energy generation process involves the conversion from *electrical to thermal energy* in a current-carrying medium (*Ohmic*, or *resistance*, or *Joule heating*). The rate at which energy is generated by passing a current I through a medium of electrical resistance R_e is

$$\dot{E}_g = I^2 R_e \quad (3.37)$$

If this power generation (W) occurs uniformly throughout the medium of volume V , the volumetric generation rate (W/m³) is then

$$\dot{q} \equiv \frac{\dot{E}_g}{V} = \frac{I^2 R_e}{V} \quad (3.38)$$

Energy generation may also occur as a result of the deceleration and absorption of neutrons in the fuel element of a nuclear reactor or exothermic chemical reactions occurring within a medium. Endothermic reactions would, of course, have the inverse effect (a thermal energy sink) of converting thermal energy to chemical bonding energy. Finally, a conversion from electromagnetic to thermal energy may occur due to the absorption of radiation within the medium. The process occurs, for example, when gamma rays are absorbed in external nuclear reactor components (cladding,

thermal shields, pressure vessels, etc.) or when visible radiation is absorbed in a semitransparent medium. Remember not to confuse energy generation with energy storage (Section 1.3.1).

3.5.1 The Plane Wall

Consider the plane wall of Figure 3.9a, in which there is *uniform* energy generation per unit volume ($\dot{q}$ is constant) and the surfaces are maintained at $T_{s,1}$ and $T_{s,2}$. For constant thermal conductivity k , the appropriate form of the heat equation, Equation 2.20, is

$$\frac{d^2T}{dx^2} + \frac{\dot{q}}{k} = 0 \quad (3.39)$$

The general solution is

$$T = -\frac{\dot{q}}{2k}x^2 + C_1x + C_2 \quad (3.40)$$

where C_1 and C_2 are the constants of integration. For the prescribed boundary conditions,

$$T(-L) = T_{s,1} \quad \text{and} \quad T(L) = T_{s,2}$$

The constants may be evaluated and are of the form

$$C_1 = \frac{T_{s,2} - T_{s,1}}{2L} \quad \text{and} \quad C_2 = \frac{\dot{q}}{2k}L^2 + \frac{T_{s,1} + T_{s,2}}{2}$$

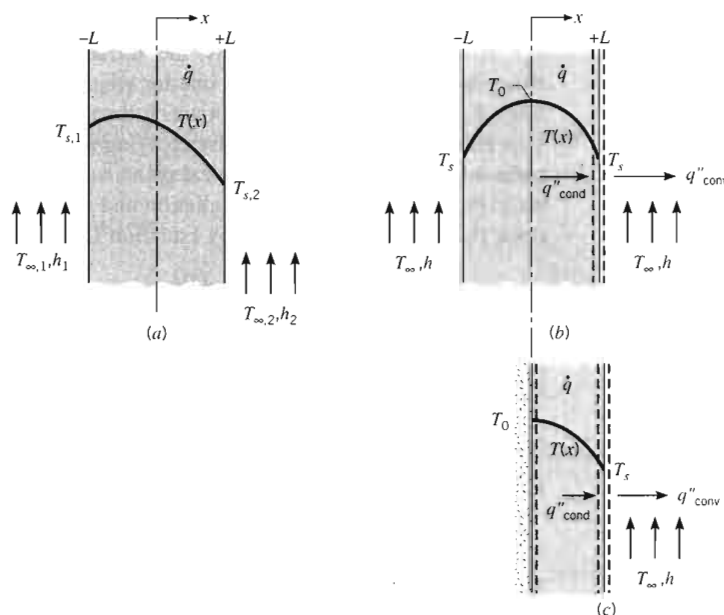


FIGURE 3.9 Conduction in a plane wall with uniform heat generation. (a) Asymmetrical boundary conditions. (b) Symmetrical boundary conditions. (c) Adiabatic surface at midplane.

in which case the temperature distribution is

$$T(x) = \frac{\dot{q}L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \frac{T_{s,2} - T_{s,1}}{2} \frac{x}{L} + \frac{T_{s,1} + T_{s,2}}{2} \quad (3.41)$$

The heat flux at any point in the wall may, of course, be determined by using Equation 3.41 with Fourier's law. Note, however, that *with generation the heat flux is no longer independent of x .*

The preceding result simplifies when both surfaces are maintained at a common temperature, $T_{s,1} = T_{s,2} \equiv T_s$. The temperature distribution is then *symmetrical* about the midplane, Figure 3.9b, and is given by

$$T(x) = \frac{\dot{q}L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + T_s \quad (3.42)$$

The maximum temperature exists at the midplane

$$T(0) \equiv T_0 = \frac{\dot{q}L^2}{2k} + T_s \quad (3.43)$$

in which case the temperature distribution, Equation 3.42, may be expressed as

$$\frac{T(x) - T_0}{T_s - T_0} = \left(\frac{x}{L} \right)^2 \quad (3.44)$$

It is important to note that at the plane of symmetry in Figure 3.9b, the temperature gradient is zero, $(dT/dx)_{x=0} = 0$. Accordingly, there is no heat transfer across this plane, and it may be represented by the *adiabatic* surface shown in Figure 3.9c. One implication of this result is that Equation 3.42 also applies to plane walls that are perfectly insulated on one side ($x = 0$) and maintained at a fixed temperature T_s on the other side ($x = L$).

To use the foregoing results, the surface temperature(s) T_s must be known. However, a common situation is one for which it is the temperature of an adjoining fluid, T_∞ , and not T_s , which is known. It then becomes necessary to relate T_s to T_∞ . This relation may be developed by applying a surface energy balance. Consider the surface at $x = L$ for the symmetrical plane wall (Figure 3.9b) or the insulated plane wall (Figure 3.9c). Neglecting radiation and substituting the appropriate rate equations, the energy balance given by Equation 1.12 reduces to

$$-k \frac{dT}{dx} \Big|_{x=L} = h(T_s - T_\infty) \quad (3.45)$$

Substituting from Equation 3.42 to obtain the temperature gradient at $x = L$, it follows that

$$T_s = T_\infty + \frac{\dot{q}L}{h} \quad (3.46)$$

Hence T_s may be computed from knowledge of T_∞ , $\dot{q}$, L , and h .

Equation 3.46 may also be obtained by applying an *overall* energy balance to the plane wall of Figure 3.9b or 3.9c. For example, relative to a control surface about the wall of Figure 3.9c, the rate at which energy is generated within the wall must be balanced by the rate at which energy leaves via convection at the boundary. Equation 1.11c reduces to

$$\dot{E}_g = \dot{E}_{\text{out}} \quad (3.47)$$

or, for a unit surface area,

$$\dot{q}L = h(T_s - T_\infty) \quad (3.48)$$

Solving for T_s , Equation 3.46 is obtained.

Equation 3.46 may be combined with Equation 3.42 to eliminate T_s from the temperature distribution, which is then expressed in terms of the known quantities $\dot{q}$, L , k , h , and T_∞ . The same result may be obtained directly by using Equation 3.45 as a boundary condition to evaluate the constants of integration appearing in Equation 3.40.

EXAMPLE 3.7

A plane wall is a composite of two materials, A and B. The wall of material A has uniform heat generation $\dot{q} = 1.5 \times 10^6 \text{ W/m}^3$, $k_A = 75 \text{ W/m} \cdot \text{K}$, and thickness $L_A = 50 \text{ mm}$. The wall material B has no generation with $k_B = 150 \text{ W/m} \cdot \text{K}$ and thickness $L_B = 20 \text{ mm}$. The inner surface of material A is well insulated, while the outer surface of material B is cooled by a water stream with $T_\infty = 30^\circ\text{C}$ and $h = 1000 \text{ W/m}^2 \cdot \text{K}$.

1. Sketch the temperature distribution that exists in the composite under steady-state conditions.
2. Determine the temperature T_0 of the insulated surface and the temperature T_2 of the cooled surface.

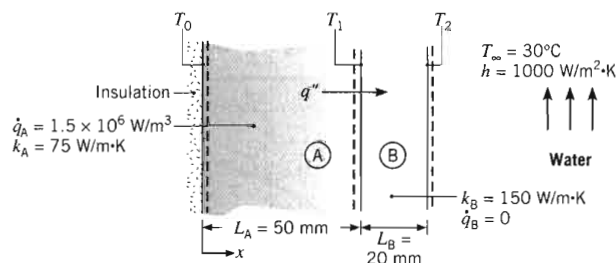
SOLUTION

Known: Plane wall of material A with internal heat generation is insulated on one side and bounded by a second wall of material B, which is without heat generation and is subjected to convection cooling.

Find:

1. Sketch of steady-state temperature distribution in the composite.
2. Inner and outer surface temperatures of the composite.

Schematic:



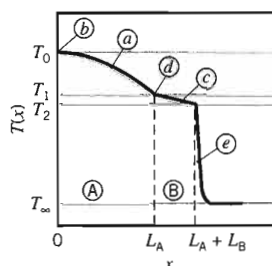
Assumptions:

1. Steady-state conditions.
2. One-dimensional conduction in x -direction.
3. Negligible contact resistance between walls.

4. Inner surface of A adiabatic.
5. Constant properties for materials A and B.

Analysis:

1. From the prescribed physical conditions, the temperature distribution in the composite is known to have the following features, as shown:
 - (a) Parabolic in material A.
 - (b) Zero slope at insulated boundary.
 - (c) Linear in material B.
 - (d) Slope change $= k_B/k_A = 2$ at interface.
 The temperature distribution in the water is characterized by
 - (e) Large gradients near the surface.



2. The outer surface temperature T_2 may be obtained by performing an energy balance on a control volume about material B. Since there is no generation in this material, it follows that, for steady-state conditions and a unit surface area, the heat flux into the material at $x = L_A$ must equal the heat flux from the material due to convection at $x = L_A + L_B$. Hence

$$q'' = h(T_2 - T_\infty) \quad (1)$$

The heat flux q'' may be determined by performing a second energy balance on a control volume about material A. In particular, since the surface at $x = 0$ is adiabatic, there is no inflow and the rate at which energy is generated must equal the outflow. Accordingly, for a unit surface area,

$$\dot{q}L_A = q'' \quad (2)$$

Combining Equations 1 and 2, the outer surface temperature is

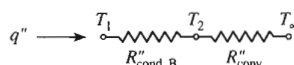
$$T_2 = T_\infty + \frac{\dot{q}L_A}{h}$$

$$T_2 = 30^\circ\text{C} + \frac{1.5 \times 10^6 \text{ W/m}^3 \times 0.05 \text{ m}}{1000 \text{ W/m}^2 \cdot \text{K}} = 105^\circ\text{C} \quad \triangleleft$$

From Equation 3.43 the temperature at the insulated surface is

$$T_0 = \frac{\dot{q}L_A^2}{2k_A} + T_1 \quad (3)$$

where T_1 may be obtained from the following thermal circuit:



That is,

$$T_1 = T_\infty + (R''_{\text{cond,B}} + R''_{\text{conv}}) q''$$

where the resistances for a unit surface area are

$$R''_{\text{cond,B}} = \frac{L_B}{k_B} \quad R''_{\text{conv}} = \frac{1}{h}$$

Hence,

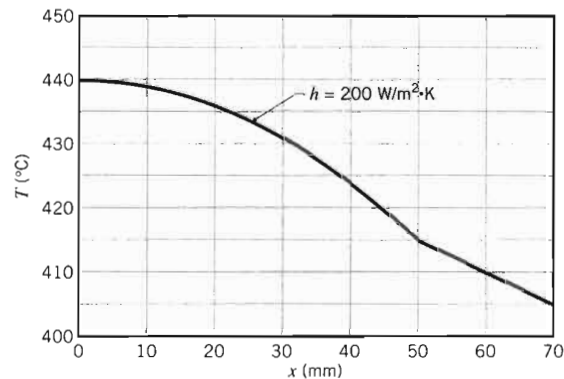
$$\begin{aligned} T_1 &= 30^\circ\text{C} + \left(\frac{0.02 \text{ m}}{150 \text{ W/m} \cdot \text{K}} + \frac{1}{1000 \text{ W/m}^2 \cdot \text{K}} \right) \\ &\quad \times 1.5 \times 10^6 \text{ W/m}^3 \times 0.05 \text{ m} \\ T_1 &= 30^\circ\text{C} + 85^\circ\text{C} = 115^\circ\text{C} \end{aligned}$$

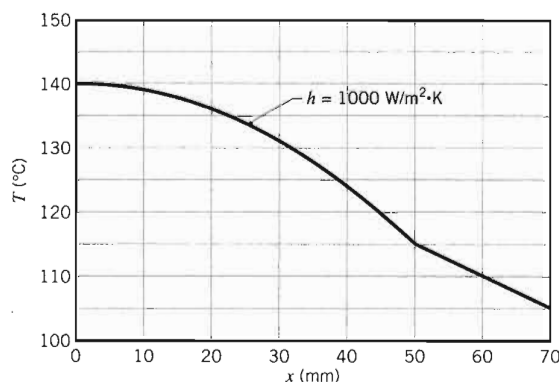
Substituting into Equation 3,

$$\begin{aligned} T_0 &= \frac{1.5 \times 10^6 \text{ W/m}^3 (0.05 \text{ m})^2}{2 \times 75 \text{ W/m} \cdot \text{K}} + 115^\circ\text{C} \\ T_0 &= 25^\circ\text{C} + 115^\circ\text{C} = 140^\circ\text{C} \end{aligned}$$

Comments:

1. Material A, having heat generation, cannot be represented by a thermal circuit element.
2. Since the resistance to heat transfer by convection is significantly larger than that due to conduction in material B, $R''_{\text{conv}}/R''_{\text{cond}} = 7.5$, the surface-to-fluid temperature difference is much larger than the temperature drop across material B, $(T_2 - T_\infty)/(T_1 - T_2) = 7.5$. This result is consistent with the temperature distribution plotted in part 1.
3. The surface and interface temperatures (T_0 , T_1 , and T_2) depend on the generation rate $\dot{q}$, the thermal conductivities k_A and k_B , and the convection coefficient h . Each material will have a maximum allowable operating temperature, which must not be exceeded if thermal failure of the system is to be avoided. We explore the effect of one of these parameters by computing and plotting temperature distributions for values of $h = 200$ and $1000 \text{ W/m}^2 \cdot \text{K}$, which would be representative of air and liquid cooling, respectively.





For $h = 200 \text{ W/m}^2 \cdot \text{K}$, there is a significant increase in temperature throughout the system and, depending on the selection of materials, thermal failure could be a problem. Note the slight discontinuity in the temperature gradient, dT/dx , at $x = 50 \text{ mm}$. What is the physical basis for this discontinuity? We have assumed negligible contact resistance at this location. What would be the effect of such a resistance on the temperature distribution throughout the system? Sketch a representative distribution. What would be the effect on the temperature distribution of an increase in $\dot{q}$, k_A , or k_B ? Qualitatively sketch the effect of such changes on the temperature distribution.

4. This example is provided as two ready-to-solve models in *IHT*, which may be accessed in *Examples* on the *menu bar*. The first approach uses the model builder, *Models/1-D, Steady-State Conduction*, which solves for temperature distributions and heat rates in plane walls, cylinders, and spheres. The second approach demonstrates how to represent temperature distributions as *User-Defined Functions*, which for this situation represents two piece-wise distributions in materials A (quadratic) and B (linear).

3.5.2 Radial Systems

Heat generation may occur in a variety of radial geometries. Consider the long, solid cylinder of Figure 3.10, which could represent a current-carrying wire or a fuel

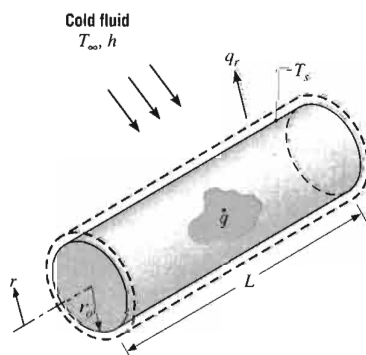


FIGURE 3.10
Conduction in a solid cylinder with uniform heat generation.